

Theory:

An RC circuit exhibits exponential behavior during charging and discharging:

Charging Equation:

$$V_c(t) = V_{in} (1 - e^{-\frac{t}{RC}})$$

Discharging Equation:

$$V_c(t) = V_0 \cdot e^{-\frac{t}{RC}}$$

$V_c(t)$ = voltage across capacitor at time t

V_{in} = input voltage (step applied)

V_0 = initial capacitor voltage

R = resistance (Ω), C = capacitance (F), $\tau = RC$ = time constant.

Equipments:

- 1) Resistors: 470Ω and 1000Ω (2 pieces total)
- 2) Capacitors: $1\mu F$ (1 piece)
- 3) Bread Board
- 4) Connecting Wires.
- 5) Signal Generator
- 6) Oscilloscope

Working Diagram :

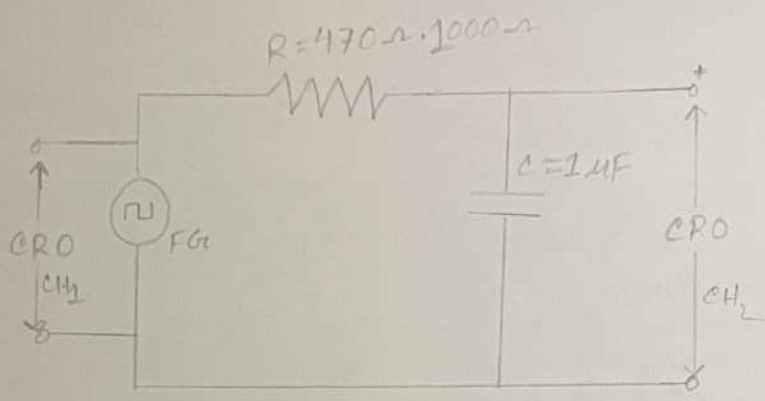
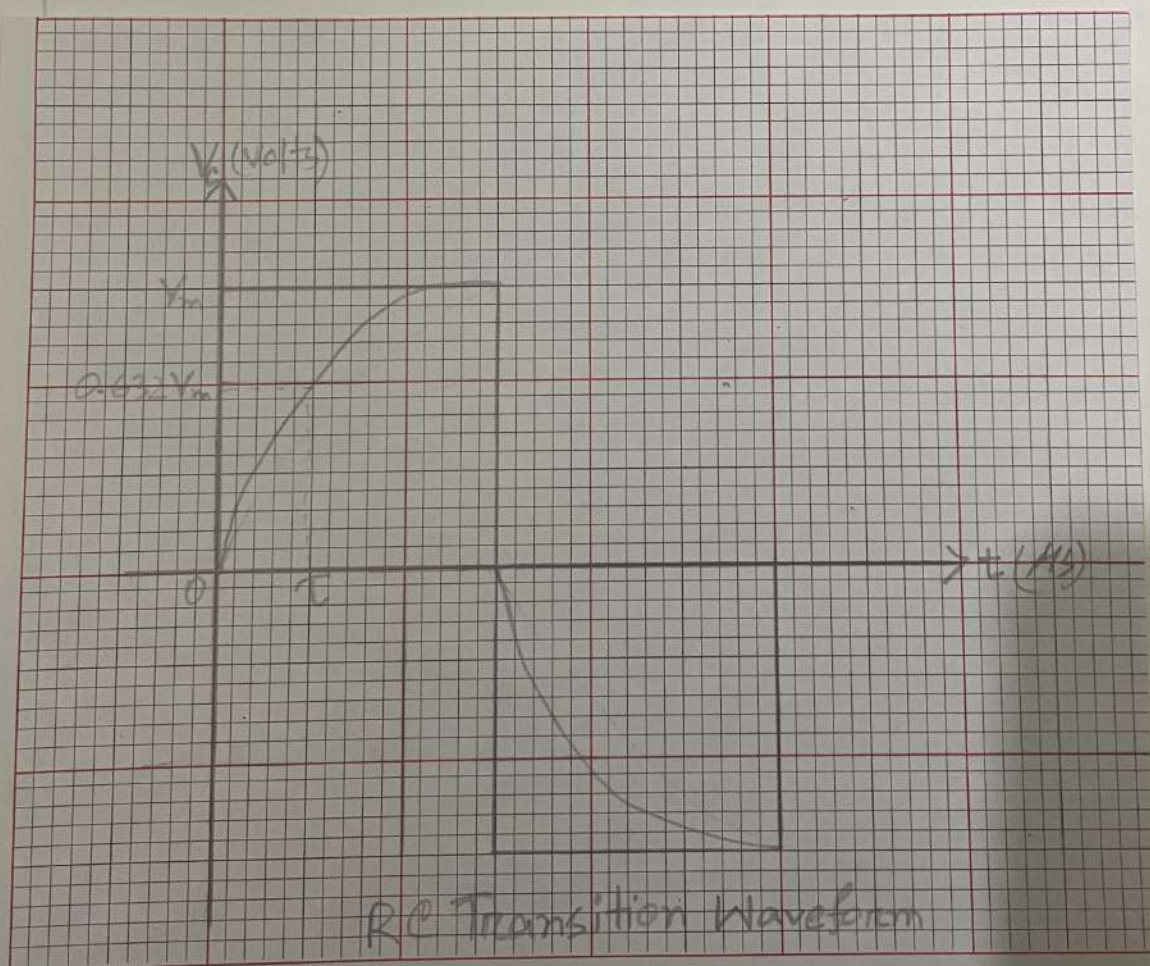


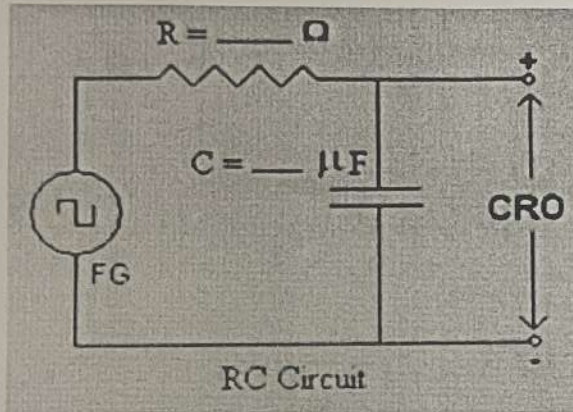
Figure: RC Circuit

Output Analysis:

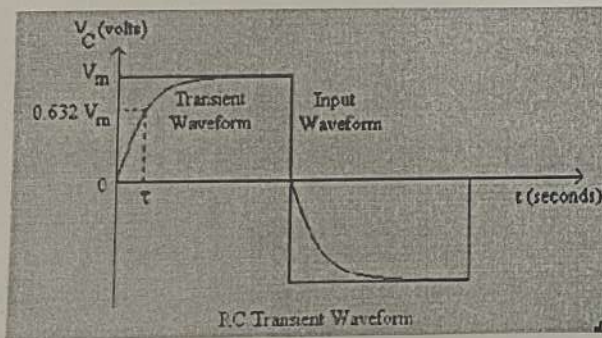


Abdullah Al Junaid
2024-3-60-082

Working Diagram:



Output analysis:



Procedure:

- Connect the circuit as per the diagram.
- Apply square wave (p-p 10 V) using a function generator.
- Set the frequency low enough (e.g., 100 Hz) so you can see the full charging/discharging waveform.
- Connect Channel 1 of the oscilloscope to input voltage (V_{in}).
- Connect Channel 2 to capacitor voltage (V_c).
- Observe and capture the charging and discharging curves from the oscilloscope.

Experimental Data:

Sl No.	Signal generator voltage (p-p)	Resistance	Capacitance	Time constant (Experimental) $\tau = 63.2\%$ of maximum output voltage (Charging curve)	Time constant (Theoretical) $\tau = RC$	% of error
1	15V	470 Ω	1 μF	420 μs	470 μs	10.63%
2	15V	1000 Ω	1 μF	900 μs	1000 μs	10%

Sub
10/12/24

Discussion:

In this experiment, the transient response of an RC circuit was observed using a square wave input. The oscilloscope showed exponential charging and discharging curves of the capacitor voltage, which matched the theoretical behavior. The time constant influenced the rate of voltage change. Small differences between theoretical and experimental results may be due to component tolerances and measurement errors.